perplexity

what frequency electromagnetic fields

Electromagnetic fields span a wide range of frequencies, making up the electromagnetic spectrum, which includes everything from very low-frequency radio waves to high-frequency gamma rays.[1] [2] [3]

Main Frequency Ranges

- **Radio waves:** Less than 3×10^9 Hz (< 3 GHz), with typical power line frequencies at 50–60 Hz.[4] [5] [3]
- **Microwaves:** From 3×10^9 Hz to 3×10^{12} Hz (3 GHz to 3 THz).[3]
- **Infrared:** From 3×10^{12} Hz to 4.3×10^{14} Hz (about 3 THz to 430 THz).[3]
- **Visible light:** From about 4.3×10^{14} Hz (red) to 7.5×10^{14} Hz (violet).[2] [3]
- **Ultraviolet:** From 7.5×10^{14} Hz to 3×10^{16} Hz.[3]
- **X-rays:** From 3×10^{16} Hz to 3×10^{19} Hz.[3]
- **Gamma rays:** Above 3×10^{19} Hz.[3]

Examples

- **Power lines and electrical appliances:** Extremely Low Frequency (ELF), usually 50 Hz or 60 Hz.[5] [4]
- **Microwave ovens:** Commonly operate at 900 MHz and 2.56 GHz.[2]
- **TV and radio broadcast:** Typically use frequencies from kilohertz (kHz) to hundreds of megahertz (MHz).[1] [3]

In summary, electromagnetic fields exist at frequencies from nearly zero hertz up to more than 10^{20} Hz, covering a vast array of applications and physical phenomena.[1] [2] [3]

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